

Particulate Matter Monitoring & Health Effects

Issue 2026-09-07 | Entry window 7 September 2026 | 20 records screened in scope

20RECORDS IN SCOPE
(17 EXCLUDED)**8**EFFECT ESTIMATES
EXTRACTED**8**WITH A HUMAN
HEALTH ENDPOINT**8**ON THE SENSING /
EXPOSURE AXIS**7**FROM JOURNALS
PUBMED OMITS

Signal of the day

A regulatory station and a mobile monitor 250 m apart disagree enough to publish a different AQI colour one time in ten — and distance does not predict when.

Paired fixed-station and simultaneous mobile PM_{2.5} measurement at three contrasting sites across the Boston metro area, spring and summer 2025 (*Environ Sci Technol*), puts a number on what a sparse network cannot see. Typical agreement is good — mean station-to-mobile difference **2.0 $\mu\text{g}/\text{m}^3$** (median 1.5, IQR 0.6–2.8) — but the tail is not: dynamic events produced discrepancies exceeding **300% in relative terms and 800 $\mu\text{g}/\text{m}^3$ in absolute terms**, and in **10.4% of cases**, rising to **21.8% at one site in spring**, the two instruments implied *different AQI categories*. The finding that matters operationally is the negative one: **Euclidean distance alone could not explain the differences**. Spatial representativeness is therefore not a radius, which is exactly the assumption embedded in every siting guideline and every inverse-distance interpolation built on top of a reference network. Relative humidity was the strongest single correlate the abstract reports ($r = -0.84$) — note the sign is against a signed centre-minus-mobile difference, not an absolute one, so it is a direction, not a magnitude, and the humidity result would need the full text to interpret. Two seasons, one metro area, and mobile measurement carries its own inlet and response-time artefacts. Even so, this is the cleanest published quantification of the gap that dense low-cost deployment exists to close, and it converts that case from a spatial-coverage argument into a public-messaging one.

What changed today

- **The sensing-axis fix worked, and it worked completely.** Yesterday's action item was to AND a particle clause into the connector's sensing query, which had returned 13 hits of which 10 were rejected as topically irrelevant. Rewritten, it returned **10 hits, all 10 particle-bearing, with zero topical false positives**. The phycotoxin, football-heading and evapotranspiration class of error is gone. Recall did not suffer: the axis still surfaced every sensing record the health axis missed.
- **Crossref-by-ISSN carried this issue — 7 of 20 records, from journals PubMed structurally cannot index.** Yesterday the same leg returned 11 raw and 0 in scope. Today it returned 46 raw and 7 in scope, including the signal of the day and three *Atmos Chem Phys* papers. This is the largest contribution this leg has made, and it is a direct vindication of keeping it: *ACP* and *AMT* are invisible to the PubMed spine, so a PubMed-only watch would have shipped 13 records and missed the one that matters most to this project.
- **Eleven of twenty records carry no human health endpoint — the inverse of yesterday.** 6 September was 10 of 15 *with* an endpoint. Today's issue is dominated by instrumentation and atmospheric process work: a UAV single-particle sampler, a tethered balloon, two dust emission schemes, a forecasting network and a bioaerosol sampler validation. Read this as a composition swing driven by which source leg fired, not as a shift in the field.
- **The trial-watch zero is a US federal holiday, not a weekend.** ClinicalTrials.gov returned **0 studies updated registry-wide** on 7 September — confirmed with a no-term control. 7 September 2026 is **Labor Day**. The prior reading was that registry zeros are a weekend artefact; this extends it to federal holidays and rules out the weekend framing as the complete explanation. `trials.json` is unchanged at 11 entries.
- **The forest plot is thin and structurally narrow: 8 estimates, 2 papers.** Three are myopia prevalence ratios, five are AECOPD incidence rate ratios, and three of those five are not particulate at all (ozone twice, diurnal temperature range once). The panel shows within-study precision, not between-study agreement, and is captioned as such.
- **Sulfate, not mass, carries the myopia signal in 212,566 children.** Tianjin grades 4–6, machine-learning exposure surfaces at school locations: SO_4^{2-} top vs bottom quartile gives **PR 1.10 (1.07–1.13)** while the whole-mixture qgcomp estimate is only **1.017 (1.007–1.027)** per quartile. A component-resolved contrast that survives when unspesiated mass barely moves is the recurring argument against mass-only networks — and school-location assignment, not residence, is the design's own main threat.
- **Three pipeline defects closed in the same run.** The DOI gate gained an UNREGISTERED-DOI warn class after an *Environ Res* record FAILED

with a DOI that PubMed asserts but Elsevier has not yet deposited (Crossref *and* the Handle API both 404). `plots.py` gained 17 new design labels — the entire issue had fallen through to *Other / mixed* — and the geography map now resolves the labels that were silently pooling into the *Global / multi-region* fallback, which had swallowed the only **Sub-Saharan Africa** and only **Latin America** records in the window.

Corpus analytics

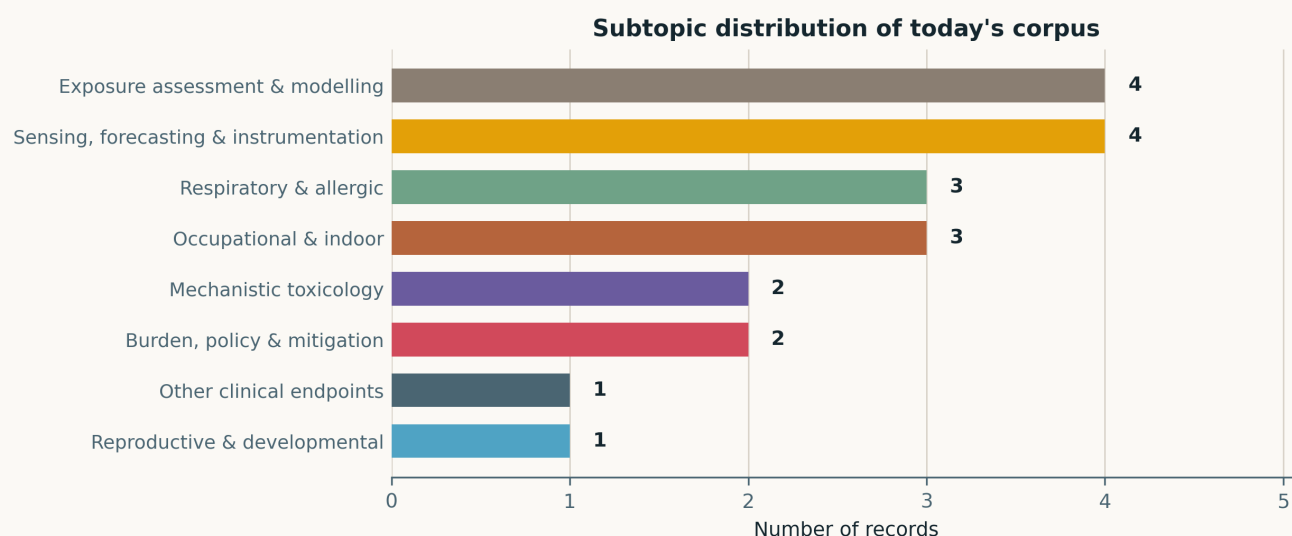


Fig. Subtopic assignment of the 20 in-scope records. Eight of the ten clusters are occupied; *Cardiovascular & metabolic* and *Neuro / mental health* are empty for the first time in this archive's recent history. *Exposure assessment* and *Sensing, forecasting & instrumentation* tie at four each and together take 8 of 20 — the methodological half of the watch outweighing the epidemiological half, which is unusual and is a source-leg effect.

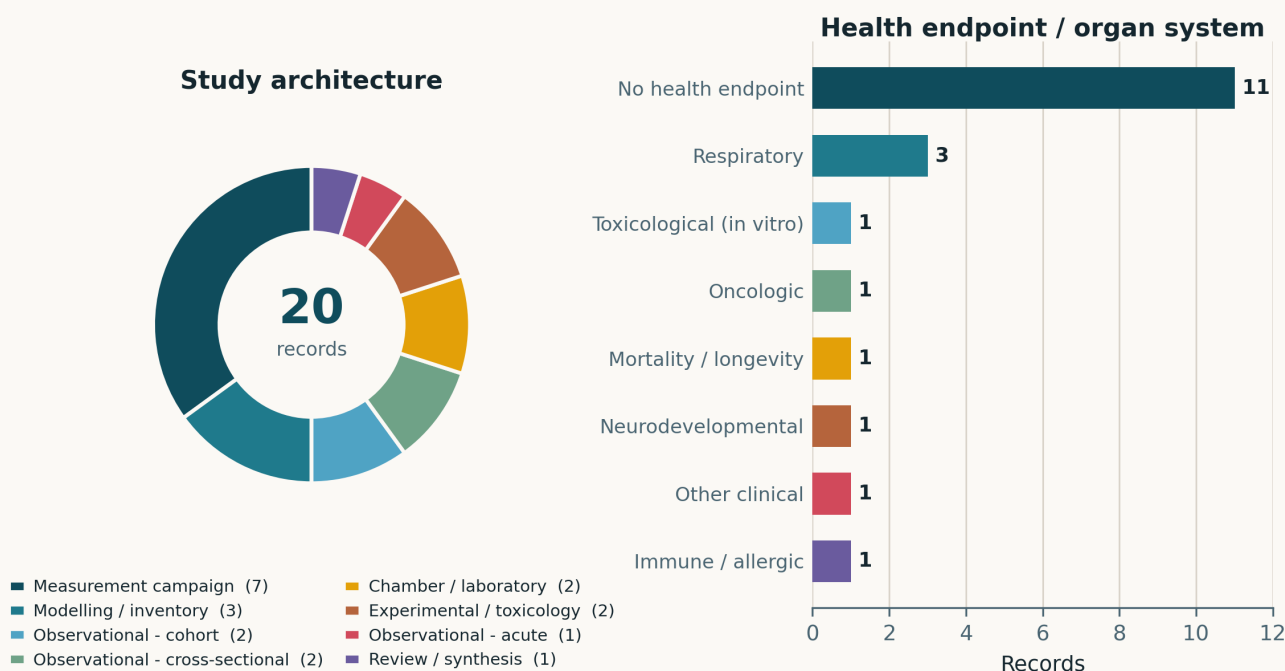


Fig. Left — study architecture. **Measurement campaigns dominate at 7 of 20**, against 5 for observational designs of all kinds combined. The campaign bucket is broad by construction: it holds a UAV deployment, a tethered balloon, a citizen-science snow and dust survey, a mobile-versus-fixed co-measurement, a six-city indoor survey, an event-based bioaerosol watch and a hybrid low-cost/BAM network. None of these is a sensor co-location study, which is a distinct and stricter claim. Right — health endpoint. **Eleven of 20 carry no health endpoint at all**; of the nine that do, one is in-vitro toxicology, so the human-endpoint count is eight.

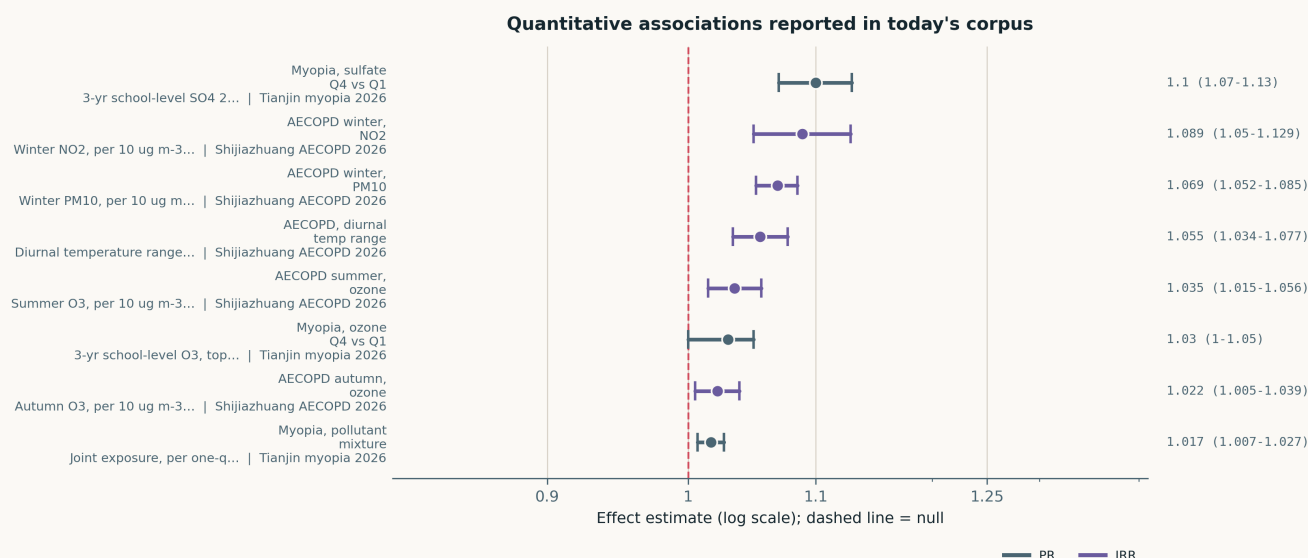


Fig. All eight quantitative estimates in the corpus, on a common logarithmic axis. They come from only two papers and the contrasts are not mutually comparable — three are prevalence ratios for myopia against long-term quartile contrasts of a single component, of a mixture and of ozone, and five are incidence rate ratios for COPD exacerbation against 7-day mean lag0–6 windows. Three of the eight are not particulate: ozone in summer and autumn, and a per-1°C diurnal temperature range. Read direction and precision only; the visual clustering here is within-study, not corroboration.

Life-course windows addressed today (records may span several stages)

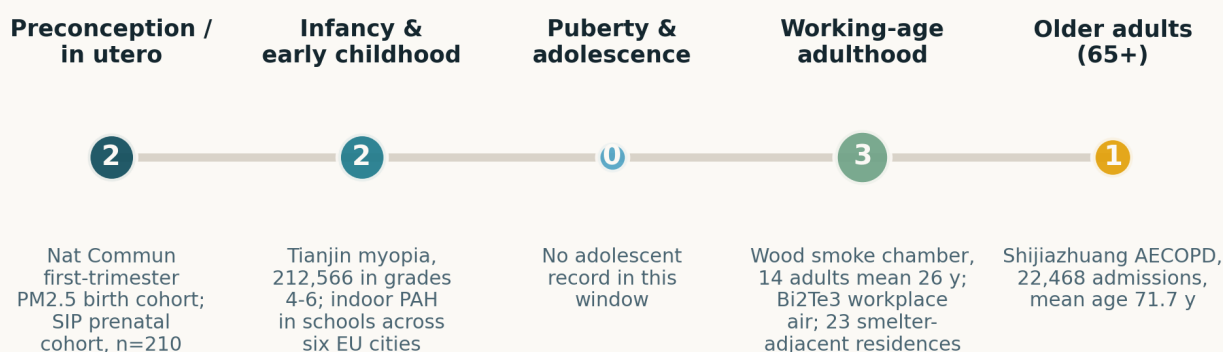


Fig. Life-course windows addressed; counts are non-exclusive and only 8 of the 20 records place anywhere on this axis, because eleven have no human endpoint. The prenatal window is occupied by both of today's cohort records; the adolescent window is empty again, as it has been for most of this archive.

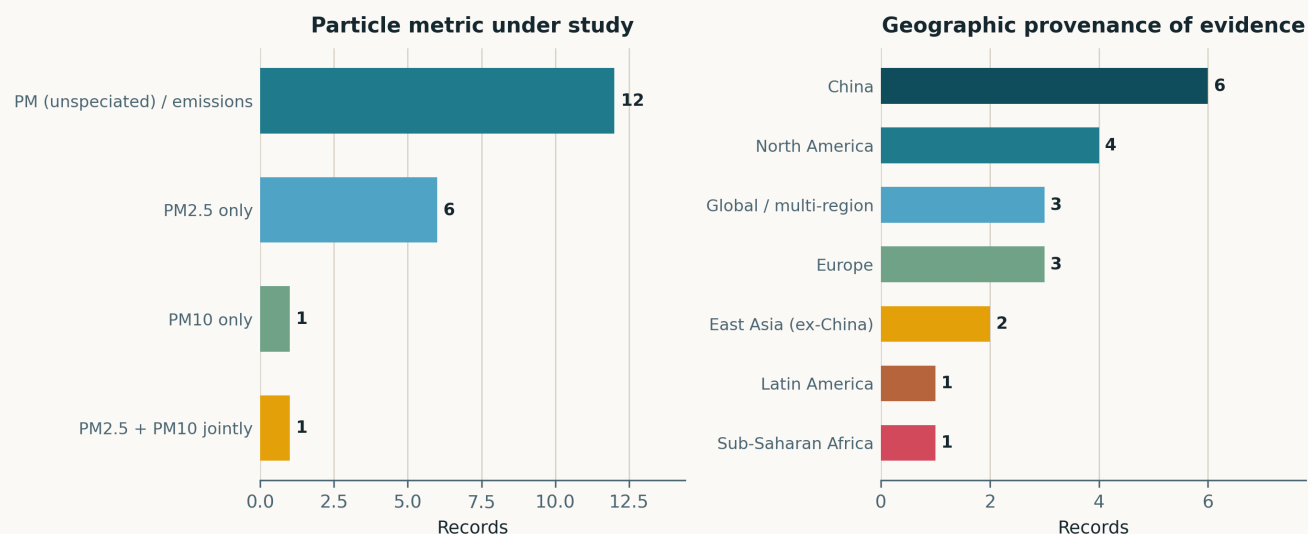


Fig. Left — particle metric. PM (unspeciated) / emissions is the largest bin at 12 of 20, and here that label is doing more work than usual: it holds modelled dust emission flux, eBC and UV-absorbing PM, single-particle composition classes, PAH mass, workplace metal aerosol and a virus-laden collection liquid. Only 8 of 20 report a size-resolved PM metric. Right — geographic provenance. China supplies 6, and Sub-Saharan Africa and Latin America appear at one record each — both were pooled into the fallback bucket until the geography map was corrected in this run.

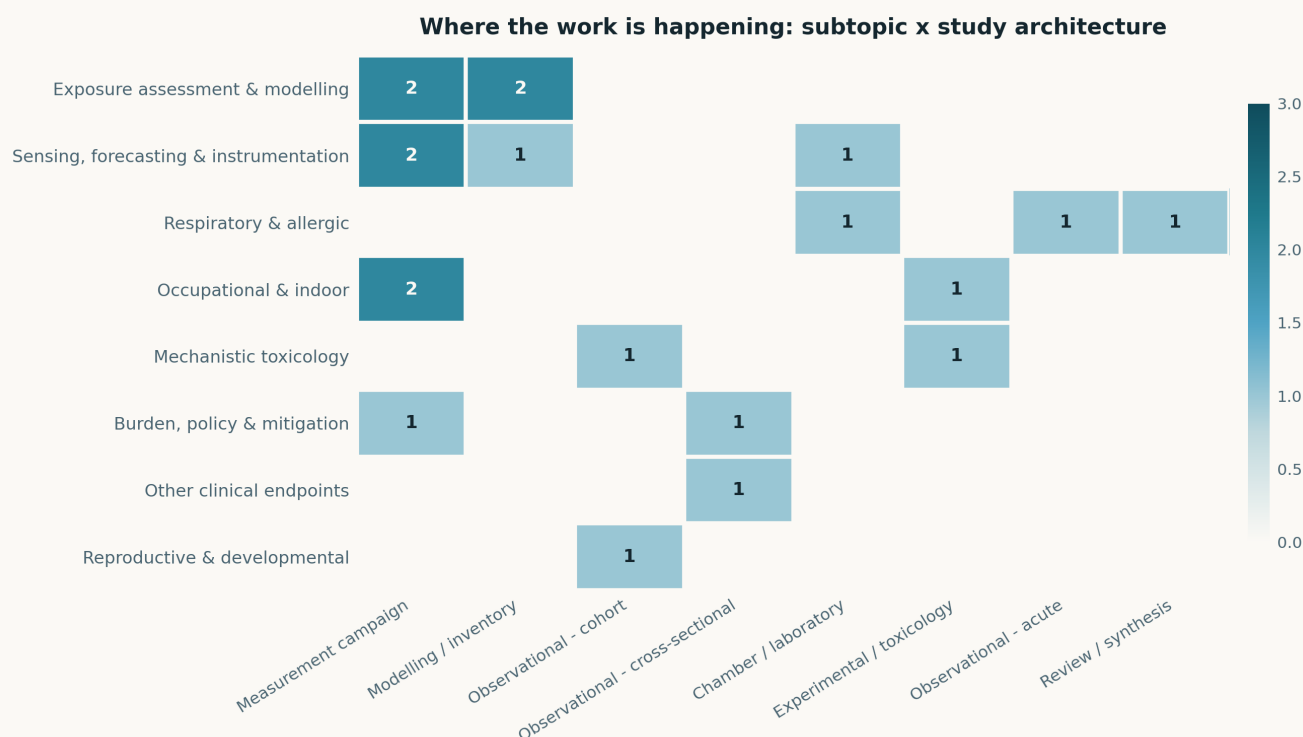


Fig. Subtopic $\times$ study-architecture cross-tabulation. **No cell holds more than two records** — four are tied at the maximum — so this is a maximally dispersed day; *Exposure assessment* splits evenly between measurement campaigns and modelling. The structural absence worth naming is that no cell links an observational design to the sensing or exposure clusters — no record this window measures particles with a novel instrument *and* attaches a health outcome to what it measured. The Addis Ababa network is the nearest miss, and it reaches its health claim through published risk coefficients rather than its own outcome data.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

4 records

Boston sparse network 2026 Environ Sci Technol

What sparse fixed air-quality networks miss, measured against mobile monitoring

Fixed-station $PM_{2.5}$ paired with simultaneous mobile measurement at a dense set of locations within a 250 m radius, at three contrasting areas of the Boston metro area across spring and summer 2025. Typical agreement is close — mean difference $2.0 \mu g/m^3$, median 1.5, IQR 0.6–2.8 — but dynamic events drove differences past 300% and $800 \mu g/m^3$, and **10.4% of cases** (21.8% at one site in spring) fell either side of an AQI category boundary. Crucially **Euclidean distance alone could not explain the discrepancies**, which undercuts radius-based representativeness and every inverse-distance product built on a reference network. Relative humidity was the strongest reported correlate ($r = -0.84$), a signed rather than absolute association that the abstract does not fully disambiguate. Two seasons and one metro area, and mobile platforms carry inlet and response-time artefacts of their own; but this is the number the dense-network argument has been missing.

doi: 10.1021/acs.est.6c03923

UAV single-particle CCSEM 2026 Environ Sci Technol

Size-resolved aerosol composition and mixing state through the boundary layer from a UAV

A microscopy-based particle sampler flown on a UAV at up to 30 m vertical and 1 min temporal resolution in a coastal boundary layer; over **100,000 particles** classified by computer-controlled SEM with EDX. The observed transition runs from sea-breeze aerosol dominated by aged sea salt and dust (46–83%) to a continentally influenced population dominated by organic-sulfur particles (57–89%), and the mixing state index falls from **53% to 27%** as those particles form — i.e. toward external mixing. Mixing state showed strong diurnal and size dependence but weak vertical structure. Offline microscopy is not a real-time measurement and the sampling is necessarily intermittent, so the “temporal evolution” is a sequence of snapshots; one coastal site, and no companion mass measurement is described. The transferable point for network design is that composition and mixing state vary on scales a fixed filter sampler averages away entirely.

doi: 10.1021/acs.est.6c03627

Bogota BiLSTM Huber 2026 Atmosphere

Asymmetric-loss BiLSTM for 24 h $PM_{2.5}$ forecasting and exceedance detection

Hourly records from the Puente Aranda industrial-corridor station, Bogotá, 2022–2025 ($n = 31,961$). The motivating statistic is sharp: **98.7%** of days exceeding the WHO 2021 $15 \mu g/m^3$ guideline contain at least one hour above 25, median peak-to-mean **1.90**, so a model trained on mean error systematically misses the episodes that matter. Three mechanisms are stacked — an asymmetric Huber loss penalising underprediction α -fold, exceedance sample weighting, and Optuna-TPE over 100 trials. The honest result is a reported **regression–detection tension**: no α minimised both global and peak MAE. Optimisation chose $\alpha = 2.80$, giving the best detection (AP **0.724**, 95% CI 0.582–0.821) and lowest peak MAE ($10.64 \mu g/m^3$) at the cost of $R^2 \approx 0$ globally; $\alpha = 1.5$ is the compromise (AP 0.718, RMSE 6.65, $R^2 = 0.372$). One station, so nothing here is a spatial claim, and the wide bootstrap intervals on AP overlap across all configurations — the ranking is weaker than the point estimates suggest.

doi: 10.3390/atmos17090871

Coriolis sampler inactivation 2026 Indoor Air*Sampler-induced loss of respiratory virus infectivity in spiked collection liquid*

Rhinovirus, SARS-CoV-2 and influenza A spiked into the collection liquid of a Coriolis μ liquid impinger and run at several airflow rates — deliberately *not* an aerosol or ambient-air collection, which isolates mechanical stress in the liquid from capture efficiency. Culturable virus fell by up to **98.6%** for SARS-CoV-2 and influenza A, while RNA copies were largely unchanged for SARS-CoV-2 and rhinovirus and reduced only for influenza A. Rhinovirus infectivity was essentially unaffected, so the effect is virus-specific rather than a general artefact. The design's strength is also its limit: it cannot say how much infectivity is lost during aerosol capture itself, only during the liquid phase, so it bounds one term of the total sampling loss. The consequence for bioaerosol surveillance is direct — a PCR-positive, culture-negative air sample is not evidence of non-infectious aerosol.

doi: 10.1155/ina/5068026

Exposure assessment & modelling**4 records****Liupan tethered eBC 2026** Atmos Chem Phys*Vertical structure of carbonaceous aerosol over the Loess Plateau from tethered balloon*

Coordinated tethered-balloon profiling of carbonaceous aerosol and meteorology at Pingliang, eastern foothills of the Liupan Mountains, during July 2023 and July 2024 campaigns. Near-surface means were **0.84 $\mu\text{g}/\text{m}^3$ eBC** and **1.24 $\mu\text{g}/\text{m}^3$ UVPM**, both decreasing with height. The substantive observation is that in the early morning and at night, when convection is weak, **UVPM aloft exceeded eBC** — attributed to nucleation from gaseous precursors rather than transport. Thermodynamic processes dominated vertical transport near the surface while enhanced dynamics aloft drove horizontal dispersion, and southerly air masses contributed disproportionately to UVPM. Two July campaigns at one site cannot separate seasonal from interannual variation, and eBC and UVPM are optical attributions from a filter-based absorption instrument, not chemical speciation — the nucleation interpretation is inference from the eBC/UVPM divergence, not a measured formation pathway.

doi: 10.5194/acp-26-12671-2026

EROD-HR/SOILHD Iberia 2026 Atmos Chem Phys*High-resolution soil erodibility and its effect on WRF-Chem dust and PM_{10}*

Two erodibility datasets tested in WRF-Chem with GOCART: EROD-HR, which folds fine-resolution topography in at 0.0625° globally and 1 km over Iberia, and SOILHD, which adds high-resolution sand/silt/clay texture and removes misclassified bare-soil areas at 1 km globally. Five dust episodes spanning 2022–2025 cover both local emission and long-range North African intrusion, evaluated against **PM_{10} from the SINQLAIR network in Murcia**. Both datasets improved the magnitude and timing of modelled PM_{10} peaks at inland stations; **improvements were limited at coastal stations**, which the paper does not fully resolve and which is the most interesting part of the result — coastal sites mix dust with sea salt and are the ones regulatory exceedance attribution most often disputes. Five episodes is a small evaluation set, and network PM_{10} cannot separate dust from co-occurring anthropogenic mass.

doi: 10.5194/acp-26-12641-2026

Taiwan LRT bioaerosol 2026 Environ Res*Long-range transport effects on airborne fungal concentration and persistence*

Daily wintertime fungal monitoring in northern Taiwan, comparing periods before, during and after Asian dust storms and frontal pollution events. Long-range transport raised fungal concentrations by **57–193% during events** and **17–69% in the post-event window**, with elevation persisting **one to four days** after passage. Frontal pollution enhanced concentrations more broadly than dust events did, and the assemblages differed by regime: dust storms enriched Basidiospores and *Cladosporium*, frontal events enriched *Aspergillus/Penicillium* and Ascospores. The persistence result is the useful one, because event-triggered sampling designs that stop at event end will systematically under-count exposure. Weaknesses are the usual ones for this design: event periods are defined meteorologically and confounded with the temperature, humidity and mixing changes that accompany them, and no source apportionment separates transported spores from locally resuspended ones.

doi: 10.1016/j.envres.2026.125640

St Elias proglacial dust 2026 Atmos Chem Phys*Sub-kilometre modelling of mineral dust emission from proglacial valleys*

A sub-km WRF-Chem implementation built to resolve dust emission from the proglacial valleys of the St. Elias Mountains, Canada — sources absent from current emission modelling and unaccounted for in global inventories. Land-surface inputs were rebuilt with explicit treatment of glaciofluvial deposit heterogeneity and inundation state; 5–19 d simulations across 2019–2022 were evaluated against in situ meteorology, dust measurements, camera stations and surface Doppler LiDAR. Total emission is estimated at **$1.0 \times 10^4 \text{ kg km}^{-2} \text{ d}^{-1}$** from erodible deposits over 47 simulated days, offered explicitly as a first-order figure. Skill in reproducing surface meteorology and in-valley vertical dispersion is **season-dependent**, and the emission scheme remains sensitive to soil texture, so the flux estimate inherits that sensitivity. High-latitude dust is a genuinely under-measured PM source and this is a careful first quantification, but 47 days at one mountain range does not support extrapolation.

doi: 10.5194/acp-26-12613-2026

Respiratory & allergic**3 records****Shijiazhuang AECOPD 2026** COPD*Seasonal and short-term environmental associations with COPD exacerbation admissions*

City-wide retrospective time-series of **22,468 AECOPD admissions**, 2020–2024, from the municipal medical insurance database covering all tertiary, secondary and community hospitals — unusually complete ascertainment, though ED discharges without admission are excluded. Mean age 71.7 (SD 9.1), male-to-female 3.7:1, winter peak 6,009 against summer 5,051. On the 7-day mean of lags 0–6, each **$10 \mu\text{g}/\text{m}^3$** of winter PM_{10} gave **IRR 1.069 (1.052–1.085)** and **NO_2 1.089 (1.050–1.129)**; ozone was positive in summer **1.035 (1.015–1.056)** and autumn **1.022 (1.005–1.039)**; a **1°C** wider diurnal range gave **1.055 (1.034–1.077)**. The authors are explicit that the many lag $\times$ pollutant $\times$ season tests are exploratory and that the prediction models are not a deployable early-warning system — a welcome discipline, and the reason the apparent protective multipollutant IRRs should not be read as findings. Collinear pollutants in a single city cannot be separated by adjustment.

doi: 10.1080/15412555.2026.2723102

Umea wood smoke chamber 2026 Environ Res*Controlled wood smoke exposure, sensory irritation and small airway mechanics*

Randomised, double-blind crossover chamber study: **14** healthy non-smokers (11 M, 3 F, mean age 26) exposed for 2 h to wood smoke at **PM₁ 453 ± 13 µg/m³** and to filtered air, three-week washout, with forced oscillometry (tremoFlo) alongside spirometry and Borg symptom scoring. Wood smoke produced significant **nasal irritation** ($p = 0.003$) with trends for cough and odour, but **no exposure-specific change** in any spirometric or oscillometric resistance parameter (R5, R5–R19, R5–R20). Reactance and resonant frequency moved with time irrespective of exposure, which is the control the design exists to provide. With $n = 14$ healthy young adults this is badly underpowered for small-airway effects and the null should be read as such, not as evidence of no effect — the authors say as much. The methodological point stands independently: irritative symptoms preceded any mechanical signal, so spirometry-only designs will under-detect early response.

doi: 10.1016/j.envres.2026.125629

Disaster respiratory review 2026 Thorac Res Pract*Respiratory emergencies in disasters*

Narrative review spanning earthquake silica and asbestos dust, wildfire fine particulate, flood-related mould, volcanic ash and acidic gases, chemical release, blast barotrauma and pandemic threats, synthesised from PubMed/MEDLINE, Web of Science and Scopus with EM-DAT, WHO and UNDRR reports. It reports that acute respiratory infection is a leading cause of disaster-related morbidity and mortality across age groups, that chronic respiratory patients face elevated exacerbation risk, and that ARDS management remains hardest in resource-limited settings. There is **no stated search protocol, no inclusion criteria and no risk-of-bias appraisal**, so it indexes the literature rather than weighing it, and no quantitative estimate here can be traced to a defined evidence base. Carried as a citation anchor for the disaster-PM framing and for its explicit treatment of climate change as a driver of rising burden — not as evidence.

doi: 10.4274/ThoracResPract.2026.2026-6-13

Occupational & indoor exposure**3 records****Europe indoor PAH children 2026** Environ Res*Indoor-air PAH levels and child cancer risk across six European cities*

Twelve EPA PAHs in air and deposited dust from schools, homes and sports halls in Athens, Barcelona, Colchester, Copenhagen, Helsinki and Lisbon — four of the 16 target compounds were dropped for analytical limitations, which is stated. Spatial variability is large: medians above **50,000 pg/m³** where intensive residential wood burning and poor ventilation coincide, against **under 10,000 pg/m³** in south-western Europe. High-molecular-weight PAHs ran higher in schools than homes, consistent with occupancy and accumulation of outdoor-derived particles, while low-molecular-weight species tracked household practices. Estimating air concentrations from deposited dust **underestimated HMW-PAHs**, which is a directly useful negative result for anyone tempted to substitute dust for active sampling. Excess cancer risk over a five-year primary-school exposure period spanned **10⁻⁸ to 10⁻⁵**, below the 10⁻⁴ reference. Cross-sectional sampling cannot capture seasonal wood-burning cycles.

doi: 10.1016/j.envres.2026.125643

Rouyn-Noranda citizen dust 2026 Environ Monit Assess*Citizen-science measurement of smelter-derived metals in snow and indoor dust*

Residents of **23 homes** 540–2,080 m from the Horne Smelter — Canada's only copper smelter — collected their own indoor dust and adjacent snow samples for Cu, Pb, Cd, As, Zn and Cr. Accounting for distance and wind direction, smelter location explained **73% of snow variance for Cu and Cr**, 58% for Cd and Pb, 56% for As and 44% for Zn ($p < 0.01$); for indoor dust the distance-only explanatory power was much lower at **36% (Cu) to 15% (As)**. Copper served as a smelter tracer and correlated strongly with the other metals in snow ($R^2 > 70\%$). Few residential characteristics predicted indoor dust once distance was controlled. Twenty-three homes is a small sample, snow and settled dust are deposition integrals rather than airborne concentrations, and participant-collected samples carry uncontrolled handling variance. The value is the demonstration itself: a participatory design produced a defensible source-attribution gradient in a community with a documented exposure dispute.

doi: 10.1007/s10661-026-15900-2

Bi/Bi₂Te₃ workplace ICP-OES 2026 Zhonghua Lao Dong Wei Sheng Zhi Ye Bing Za Zhi*ICP-OES determination of bismuth and bismuth telluride in workplace air*

Method development for an occupational aerosol with no established Chinese workplace standard: mixed cellulose ester filter sampling, ultra microwave digestion at 130°C for 10 minutes, then simultaneous Bi and Te by ICP-OES with Bi₂Te₃ back-calculated from either element using on-site speciation information. Linear over **0.04–30 mg/L** ($r > 0.9990$); on a 75 L sample diluted to 50 mL the minimum detectable Bi₂Te₃ is **0.006 mg/m³** via Bi and **0.008** via Te. Digestion efficiency 99.1–100.0% (Bi) and 101.2–103.3% (Te), intra-batch precision under 0.8%, 7-day room temperature stability, and no interference from Ag, Al or As up to 10 mg/L. Validated on simulated samples rather than a field inter-laboratory comparison, and thermoelectric workplaces are the only stated application. Included as the kind of low-visibility analytical groundwork on which occupational exposure limits eventually rest.

doi: 10.3760/cma.j.cn121094-20250319-00107

Mechanistic toxicology**2 records****Fetal macrophage FCER1G 2026** Nat Commun*Prenatal PM_{2.5}, fetal macrophage epigenetic reprogramming and atopic dermatitis*

Birth-cohort association between **first-trimester PM_{2.5}** and early-childhood atopic dermatitis, carried into mechanism by integrative multi-omics including single-cell profiling of placenta and fetal immune cells. The central finding is **hypomethylation of FCER1G in fetal macrophages** driving sustained overexpression that persists across developmental stages and re-emerges in M2 macrophages of AD skin and peripheral blood; functional work links FCER1G networks to NADPH-oxidase-mediated ROS and Th2 inflammatory signalling. This is a rare case of a durable, cell-type-resolved epigenetic mediator proposed for a prenatal PM_{2.5} association rather than asserted. The exposure remains a modelled residential estimate assigned to a trimester, which is the weakest link in an otherwise strong chain — first-trimester windows are precisely where exposure misclassification is largest. The persistence claim rests on cross-sectional comparisons at different stages, not on following the same cells.

doi: 10.1038/s41467-026-76298-6

6PPD-Q AC16 p53 2026 Ecotoxical Environ Saf*Cardiotoxic mechanism of the tire-derived transformation product 6PPD-quinone*

Network toxicology and docking followed by wet-lab confirmation in human AC16 cardiomyocytes. **144 putative targets** with TP53 the top hub; IC₅₀ **13.92 μ M** at 24 h. 6PPD-Q drove ROS accumulation and **p53 Ser15 phosphorylation without raising total p53 protein or TP53 mRNA** — i.e. post-translational stress activation — alongside TRAF6 upregulation, MAPK1 phosphorylation, IL6/TNF induction, mitochondrial apoptosis (membrane potential loss, Bax/Bcl-2 rise, cleaved caspase-9 and -3) and blocked late autophagic flux (LC3-II accumulation with p62 depletion). Pifithrin- α attenuated all of it, which is the load-bearing experiment. The exposure gap is total: a micromolar dose to an immortalised cardiomyocyte line has no established relationship to any airborne 6PPD-Q concentration, and no inhalation route is demonstrated. Relevant here only because 6PPD-Q is a measurable constituent of road dust and traffic-derived PM.

doi: 10.1016/j.j.ecoenv.2026.120754

Burden, policy & mitigation**2 records****Addis Ababa hybrid net 2026** Health Serv Insights*PM_{2.5} spatiotemporal variation and monetised burden from a hybrid sensor network*

Two years of hourly PM_{2.5} (2022–2023) from a **hybrid network of seven low-cost sensors and two beta attenuation monitors** — the design this project cares about, deployed in a city with almost no prior monitoring. Site means **15–33 μ g/m³**, citywide **27**, more than fivefold the WHO guideline of 5; concentrations were higher in the wet Kiremt season, with a **07:00 diurnal peak of 38**. Hazard quotients exceeded 1 in all age groups. The cancer-risk estimate is honestly reported as unstable: ELCR exceeded 10^{-4} under a slope factor of 0.008 but stayed acceptable at 0.0012. Monetised DALYs give **USD 652.95 M** over two years, **0.25% of GDP** in each year, dominated by ischaemic heart disease. The paper does not describe co-location or humidity correction for the seven low-cost nodes, and every health figure is transferred from external risk coefficients, so the burden estimate inherits both instrument bias and transfer error.

doi: 10.1177/11786329261487535

Hohhot park PM removal 2026 Environ Pollut*Random-forest attribution of PM reduction by urban park green space*

Thirty-seven parks in Hohhot, summer daytime, reduction rates computed for **six PM size fractions**. Linear, GAM and random-forest models were compared under leave-one-park-out cross-validation and RF won, though the fit is modest and honestly reported — point-level R^2 **0.19–0.58**, park-level **0.18–0.72**. SHAP plus the derivatives of partial-dependence and ICE curves identified nine drivers and, more usefully, the **threshold ranges beyond which the marginal mitigation effect plateaus**: air temperature 22.08–33.26°C, RH 29.71–67.17%, wind speed 0.14–1.12 m/s, distance from centre 2.7–16.79 km, perimeter-area ratio 0.15–0.33, contiguity 0.24–0.48, aggregation index 78.46–86.85%. Summer daytime only, one semi-arid city, and the in-versus-out contrast cannot separate deposition from dispersion and shelter effects. The plateau structure is the transferable result; the specific thresholds are not.

doi: 10.1016/j.j.envpol.2026.129128

Other clinical endpoints**1 record****Tianjin myopia 2026** Environ Pollut*Long-term PM_{2.5} components and childhood myopia*

School-based study of **212,566** children in grades 4–6 in Tianjin, with three-year mean PM_{2.5}, its major components (SO₄²⁻, NO₃⁻, NH₄⁺, organic matter, black carbon) and ozone estimated by machine-learning exposure models and assigned to *school* locations. Restricted cubic splines and quartile-based modified Poisson models for single pollutants, quantile g-computation for the mixture. **Sulfate carried the signal**: top versus bottom quartile **PR 1.10 (1.07–1.13)**, against a weak non-monotonic ozone association **1.03 (1.00–1.05)** and a whole-mixture estimate of only **1.017 (1.007–1.027)** per quartile increase. Cross-sectional, so myopia onset timing is unknown and reverse pathways through indoor time are not excluded; exposure at school ignores home and commute, which for a three-year window is substantial misclassification. The component contrast is the reason to read it — a speciated signal that unspiciated mass does not reproduce is the standing argument against mass-only networks.

doi: 10.1016/j.j.envpol.2026.129120

Reproductive & developmental**1 record****SIP prenatal Bayley 2026** Child Neuropsychol*In utero PM_{2.5} and NO₂ and cognitive-behavioural development at 1–3 years*

Stress in Pregnancy cohort, $n = 210$, Queens, New York; Bayley-III and ABAS-3 at 18, 24 and 36 months; prenatal PM_{2.5} and NO₂ taken from public monitoring databases. The reported pattern is **sex-specific and pollutant-specific**: PM_{2.5} associated with poorer communication, social skills and gross motor function in **boys**, NO₂ with poorer communication, gross motor and social-emotional skills in **girls**. No effect estimates or intervals appear in the abstract, which is why nothing from this paper enters the forest plot. With $n = 210$ split by sex across three timepoints and multiple Bayley and ABAS subdomains, the multiplicity burden is severe and the crossed sex-by-pollutant pattern is exactly the shape that subgroup analysis produces by chance. Exposure from public monitors rather than a spatial model adds classical misclassification. Hypothesis-generating, and the authors largely frame it that way.

doi: 10.1080/09297049.2026.2723475

Corpus register

First author	Focus	Journal	Design	Metric	Region
Reproductive & developmental					
SIP prenatal Bayley	The effect of in utero exposure to air pollution and subsequent cognitive and behavioral development in children aged 1-3 years old	<i>Child Neuropsychol</i>	Prospective birth cohort	Prenatal PM _{2.5} and NO ₂ from public monitoring data	United States
Respiratory & allergic					

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Disaster respiratory review	Respiratory emergencies in disasters: a narrative review	<i>Thorac Res Pract</i>	Narrative review	No new measurement; wild-fire PM, silica and ash dust	Global / multi-region
Shijiazhuang AECOPD	Seasonal patterns and short-term environmental associations with AECOPD hospitalizations in Shijiazhuang, China, 2020-2024	<i>COPD</i>	Time-series	Daily PM _{2.5} / PM ₁₀ city means; lags 0-7 and 7-day mean	China
Umea wood smoke chamber	Acute controlled wood smoke exposure induces sensory airway irritation without detectable changes in small airway mechanics in healthy adults: a randomized crossover study	<i>Environ Res</i>	Randomised crossover chamber	Chamber PM1 453 +/- 13 ug m-3, 2 h vs filtered air	Europe
Mechanistic toxicology					
6PPD-Q AC16 p53	Exploring the targets and mechanisms of 6PPD-Q-induced cardiotoxicity: a network toxicology and molecular docking study	<i>Ecotoxicol Environ Saf</i>	In silico + in vitro	No airborne metric; 6PPD-Q is a road-dust constituent	China
Fetal macrophage FCER1G	Prenatal PM _{2.5} exposure drives epigenetic reprogramming of fetal macrophages linked to atopic dermatitis	<i>Nat Commun</i>	Birth cohort + multi-omics	First-trimester ambient PM _{2.5} , modelled residential	South Korea
Occupational & indoor					
Bi/BiTe3 workplace ICP-OES	Determination of bismuth and bismuth telluride in workplace air by ultra microwave digestion with inductively coupled plasma optical emission spectrometry	<i>Zhonghua Lao Dong Wei Sheng Zhi Ye Bing Za Zhi</i>	Analytical method validation	MCE filter, 75 L; Bi and Te by ICP-OES after digestion	China
Europe indoor PAH children	Polycyclic aromatic hydrocarbon (PAH) levels in indoor air and their associated health risks for children across Europe	<i>Environ Res</i>	Multi-city indoor survey	Active indoor air sampling plus deposited dust, 12 PAHs	Europe
Rouyn-Noranda citizen dust	Metal contamination in residential snow and indoor dust near the Horne Smelter: a citizen-science study in Rouyn-Noranda, Quebec	<i>Environ Monit Assess</i>	Citizen-science sampling	Cu, Pb, Cd, As, Zn, Cr in snow and settled indoor dust	Canada
Sensing, forecasting & instrumentation					
Bogota BiLSTM Huber	BiLSTM with asymmetric Huber loss for 24 h PM _{2.5} forecasting and early exceedance detection at an industrial monitoring station in Bogota, Colombia	<i>Atmosphere</i>	Deep-learning forecasting	Hourly PM _{2.5} , Puente Aranda industrial corridor	Colombia
Boston sparse network	Air quality in cities: what do sparse sensor networks miss?	<i>Environ Sci Technol</i>	Mobile vs fixed co-measurement	Fixed-station PM _{2.5} vs mobile within a 250 m radius	United States
Coriolis sampler inactivation	Sampler-induced loss of respiratory virus infectivity revealed by spiked collection-liquid experiments	<i>Indoor Air</i>	Laboratory sampler validation	Coriolis micro liquid impinger, spiked collection liquid	Laboratory
UAV single-particle CCSEM	Temporal evolution and vertical distribution of size-resolved aerosol composition and mixing state within the boundary layer using an unmanned aerial vehicle	<i>Environ Sci Technol</i>	UAV instrument deployment	Microscopy sampler on a UAV, 30 m and 1 min resolution	Global / multi-region
Exposure assessment & modelling					
EROD-HR/SOILHD Iberia	Impact of high-resolution soil erodibility datasets on dust simulations in WRF-Chem with the GOCART scheme	<i>Atmos Chem Phys</i>	Regional CTM evaluation	Modelled PM ₁₀ vs SINCLAIR network, 5 dust episodes	Spain
Liupan tethered eBC	Measurement report: impacts of thermodynamic and dynamic processes on the vertical distribution of carbonaceous aerosols: lessons from in-situ observations at the eastern foothills of Liupan Mountains, Loess Plateau	<i>Atmos Chem Phys</i>	Tethered-balloon profiling	eBC and UVPM profiles, tethered balloon, July 23 and 24	China
St Elias proglacial dust	Modelling mineral dust emissions from proglacial valleys of the St. Elias Mountains, Canada	<i>Atmos Chem Phys</i>	Sub-km CTM development	Modelled dust emission vs in situ, camera and Doppler	Canada
Taiwan LRT bioaerosol	Impacts of long-range transport on the concentration, composition, and persistence of airborne fungi in northern Taiwan	<i>Environ Res</i>	Event-based monitoring	Daily culturable/spore fungal counts under ADS and FP	Taiwan
Burden, policy & mitigation					
Addis Ababa hybrid net	Spatiotemporal variation of PM _{2.5} concentration and health impacts in Addis Ababa, Ethiopia: trends and patterns	<i>Health Serv Insights</i>	Network monitoring + burden	Hourly PM _{2.5} , 7 low-cost sensors plus 2 BAM, 2022-23	Ethiopia

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Hohhot park PM removal	Quantifying the reduction in particulate matter by urban park green spaces using a random forest model: a case study of Hohhot, China	<i>Environ Pollut</i>	Cross-sectional + ML	Six PM size fractions, summer daytime, in/out of park	China
Other clinical endpoints					
Tianjin myopia	Association of long-term exposure to ambient air pollution and myopia in Chinese children	<i>Environ Pollut</i>	School-based cross-sectional	3-yr PM _{2.5} and components, ML model at school point	China

Provenance and method

Entry window **7 September 2026**, a single day, run against [Date - Entry]. **PubMed** (NCBI E-utilities) returned **8 health-axis** and **6 sensing-axis** hits to the harvester; the **PubMed connector**, run independently on both axes, returned **11** and **10** for 16 unique records. The sensing axis was rewritten this run to AND a particle clause into the free-text terms, closing the action item raised on 6 September: **all 10 hits were particle-bearing and none was rejected as off-topic**, against 10 of 13 rejected yesterday. **Europe PMC** (CREATION_DATE) returned 9 raw and contributed 5 records neither PubMed leg had. **Crossref-by-ISSN** returned **46 raw**, of which 13 were retrieved in full and **7 shipped** — the largest contribution this leg has made, and the source of the signal of the day; *Atmos Chem Phys*, *Atmos Meas Tech* and *Atmosphere* are not indexed by the PubMed spine at all. **arXiv** returned a payload but nothing inside the local recency screen. **OpenAlex** returned HTTP 429, unchanged since 28 July. **Consensus** returned three hits, all far out of window (2019, 2020, 2025) — a nil for the **ninth** consecutive run. **Scholar Gateway** was queried with a narrowed single-paper question after the 6 September budget overrun; it returned within budget but did not hold the target article, so 10.1016/j.atmosenv.2026.122350 ships no summary and joins the abstract retry list. **ClinicalTrials.gov** posted **no updates registry-wide** on 7 September, verified with a no-term control; 7 September 2026 is **Labor Day**, so the registry zero is a federal-holiday artefact and not evidence about PM trial activity. `trials.json` is unchanged at 11. Forty-four raw PubMed and Europe PMC hits deduplicated to **24 unique** on PMID → DOI → normalised title; **3 had already shipped in the 6 September issue** and were dropped. Seventeen records were excluded with a logged reason — those 3 duplicates, plus aquatic and marine suspended particulate matter, gas-phase-only exposures, a CO₂ sensor, a radiative-cooling materials study, tobacco-product chemistry and plant ecotoxicology. The DOI gate passed 20 of 20 with two advisory warnings and gained a new UNREGISTERED-DOI class this run: 10.1016/j.envres.2026.125643 is absent from both Crossref and the Handle API but is asserted by PubMed's own record, so it is a publisher deposit lag rather than a wrong identifier, and **that link may not resolve until Elsevier deposits it**. Subtopic, design, particle-metric and geography labels are assigned from title, abstract and keywords; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Seventeen new design labels and several geography labels were added to the plotting maps in this run. *Global / multi-region* remains a fallback bucket that absorbs unstated settings and bench work. Effect estimates are transcribed verbatim and use heterogeneous contrasts — see the forest-plot caption. All DOI links resolve to the publisher of record. Abstracts remain the copyright of their respective publishers.